

Boss Challenge — Paper 48 (Class 7)

Motion & Time | Soil | Perimeter & Area | Algebraic Expressions

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A scooter that keeps covering equal distances in every equal slot of time is moving with what is called:
(A) non-uniform motion
(B) circular motion
(C) oscillatory motion
(D) uniform motion
2. The loose mixture of broken rock, minerals, water, air and decayed living matter that covers the land is called:
(A) air
(B) humus
(C) water
(D) soil
3. The total length of the boundary that goes all the way around a flat closed shape is called its:
(A) diameter
(B) area
(C) perimeter
(D) volume
4. In the algebraic term $7x$, the number 7 multiplying the variable x is called the:
(A) variable
(B) coefficient
(C) exponent
(D) constant
5. The basic unit used to measure time in the international (SI) system is the:
(A) second
(B) kilogram
(C) hour
(D) metre
6. The dark, decayed remains of dead plants and animals that make soil fertile is known as:
(A) bedrock
(B) clay
(C) humus
(D) gravel

7. The amount of flat surface a closed shape covers, measured in square units, is its:
- (A) area
 - (B) breadth
 - (C) height
 - (D) perimeter
8. A symbol such as x or y that can stand for different number values in algebra is called a:
- (A) coefficient
 - (B) variable
 - (C) equation
 - (D) constant
9. A simple pendulum completes one full to-and-fro swing in 2 seconds. The time for one complete swing is its:
- (A) time period
 - (B) speed
 - (C) frequency
 - (D) amplitude
10. The different horizontal layers seen when soil is cut from top to bottom together make up the:
- (A) bedrock
 - (B) water table
 - (C) soil profile
 - (D) topsoil
11. To go once around a rectangle whose length is l and whose breadth is b , the correct formula is:
- (A) $l + b$
 - (B) $4 \times l$
 - (C) $2 \times (l + b)$
 - (D) $l \times b$
12. Terms that have exactly the same variable part, such as $3x$ and $5x$, are called:
- (A) constant terms
 - (B) coefficients
 - (C) like terms
 - (D) unlike terms
13. Aarav cycles 600 metres in 50 seconds at a steady rate. Using $\text{speed} = \text{distance} / \text{time}$, his speed is:
- (A) 10 m/s
 - (B) 12 m/s
 - (C) 30 m/s
 - (D) 0.083 m/s

14. The uppermost, darkest and most fertile layer of the soil profile, rich in humus, is called the:

- (A) topsoil (A-horizon)
- (B) water table
- (C) subsoil (B-horizon)
- (D) bedrock (C-horizon and below)

15. The area of a rectangle whose length is 12 cm and breadth is 5 cm is:

- (A) 34 cm²
- (B) 60 cm²
- (C) 60 cm
- (D) 17 cm²

16. Adding the like terms $4x + 9x$ gives:

- (A) $13x$
- (B) 13
- (C) $36x$
- (D) $13x^2$

17. A square park has each side 80 m. Reeta jogs once around its boundary (the perimeter) in 200 s. Her speed is:

- (A) 1.6 m/s
- (B) 3.2 m/s
- (C) 0.4 m/s
- (D) 16 m/s

18. Soil made mostly of the largest particles, which lets water drain through very quickly, is:

- (A) clayey soil
- (B) loamy soil
- (C) sandy soil
- (D) humus

19. The perimeter of a square whose side measures 9 m is:

- (A) 36 m
- (B) 18 m
- (C) 13 m
- (D) 81 m

20. The value of the expression $2x + 5$ when $x = 3$ is:

- (A) 13
- (B) 10
- (C) 11
- (D) 16

21. On a distance-time graph, the motion of a body that stays at rest in one place is shown by a:

- (A) straight sloping line
- (B) vertical line
- (C) curved line
- (D) straight horizontal line

22. Soil made of the smallest, tightly packed particles, which holds water well but drains slowly, is:

- (A) sandy soil
- (B) gravel
- (C) clayey soil
- (D) loamy soil

23. The area of a square field whose side is 15 m is:

- (A) 225 m²
- (B) 30 m²
- (C) 60 m²
- (D) 150 m²

24. An algebraic expression that has exactly two unlike terms, such as $3x + 5$, is called a:

- (A) trinomial
- (B) monomial
- (C) constant
- (D) binomial

25. On a distance-time graph, a straight line sloping upward at a constant angle tells us the body is moving with:

- (A) zero speed
- (B) decreasing speed
- (C) increasing speed
- (D) uniform speed

26. The balanced soil - a mixture of sand, clay and silt with plenty of humus - best for growing most crops is:

- (A) pure sandy soil
- (B) gravelly soil
- (C) pure clayey soil
- (D) loamy soil

27. A rectangle and a square have the same perimeter of 40 cm. If the rectangle is 12 cm long, its breadth is:

- (A) 8 cm
- (B) 28 cm
- (C) 4 cm
- (D) 16 cm

28. Subtracting $5a$ from $12a$ gives:

- (A) $7a^2$
- (B) $17a$
- (C) 7
- (D) $7a$

29. A car's odometer measures the total distance the vehicle has travelled, while the instrument that shows its speed is the:

- (A) thermometer
- (B) barometer
- (C) speedometer
- (D) odometer

30. The slow downward seepage of water through soil is measured as its percolation rate. The faster the percolation, the:

- (A) more humus the soil must contain
- (B) colder the soil must be
- (C) smaller the gaps between soil particles
- (D) larger the gaps between soil particles

31. The area of a right triangle is $\frac{1}{2} \times \text{base} \times \text{height}$. A triangle with base 10 cm and height 6 cm has area:

- (A) 16 cm^2
- (B) 30 cm
- (C) 60 cm^2
- (D) 30 cm^2

32. A car travels at a steady speed v . The expression for the distance it covers in time t is:

- (A) $v - t$
- (B) $v \times t$
- (C) v / t
- (D) $v + t$

33. If a bus travels 240 km in 4 hours, then its average speed for the journey is:

- (A) 96 km/h
- (B) 240 km/h
- (C) 60 km/h
- (D) 48 km/h

34. In a percolation test, 200 mL of water passes through a soil sample in 20 minutes. Its percolation rate is:

- (A) 180 mL per minute
- (B) 4000 mL per minute
- (C) 10 mL per minute
- (D) 0.1 mL per minute

35. The area of a parallelogram is base $\times$ height. One with base 14 cm and height 5 cm has area:
- (A) 70 cm²
 - (B) 19 cm²
 - (C) 35 cm²
 - (D) 38 cm²
36. A soil sample's percolation lets 8 mL drain each minute. The expression for the water drained in m minutes is:
- (A) $8 - m$
 - (B) $8m$
 - (C) $m / 8$
 - (D) $8 + m$
37. Speed is usually written as a fraction of two quantities. Speed equals distance divided by:
- (A) mass
 - (B) time
 - (C) force
 - (D) area
38. Two soils are tested. Soil A lets 300 mL through in 10 min; Soil B lets 300 mL through in 30 min. The sandier soil is:
- (A) Both are equally sandy
 - (B) Neither - both are clayey
 - (C) Soil A
 - (D) Soil B
39. A rectangular garden is 20 m by 15 m. A boy runs around its boundary 3 times. The total distance he runs is:
- (A) 900 m
 - (B) 210 m
 - (C) 70 m
 - (D) 105 m
40. The perimeter of a square of side x is best written as the expression:
- (A) $2x$
 - (B) $4x$
 - (C) $x + 4$
 - (D) x^2
41. The hands of a clock, the spinning of a fan and a swinging pendulum are all examples used to measure or mark:
- (A) temperature
 - (B) distance
 - (C) time
 - (D) mass

42. The wearing away and carrying off of the fertile top layer of soil by wind or flowing water is called:

- (A) humus formation
- (B) weathering
- (C) percolation
- (D) soil erosion

43. 1 square metre equals how many square centimetres?

- (A) 1000 cm^2
- (B) 1000000 cm^2
- (C) 100 cm^2
- (D) 10000 cm^2

44. Simplifying the expression $6x + 2x - 3x$ gives:

- (A) $5x^2$
- (B) $5x$
- (C) $11x$
- (D) 5

45. A toy car moves 5 m in the first second, 9 m in the next and 14 m in the third. This motion is:

- (A) rest
- (B) oscillatory motion
- (C) non-uniform motion
- (D) uniform motion

46. Planting trees and grass on bare slopes helps reduce soil erosion mainly because plant roots:

- (A) bind the soil and hold it in place
- (B) make the soil drain faster
- (C) turn the soil into pure clay
- (D) warm the soil up

47. A square and a rectangle have equal areas of 36 cm^2 . If the rectangle is 9 cm long, its breadth is:

- (A) 6 cm
- (B) 4 cm
- (C) 27 cm
- (D) 324 cm

48. The area of a rectangle with length l and breadth b is written algebraically as:

- (A) $l + b$
- (B) $2(l + b)$
- (C) $l - b$
- (D) lb

49. The time period of a pendulum depends mainly on the length of its thread. A longer pendulum has a:
- (A) longer time period
 - (B) shorter time period
 - (C) zero time period
 - (D) time period that does not change
50. The process by which rocks are broken down over a very long time into smaller particles that form soil is called:
- (A) evaporation
 - (B) erosion
 - (C) percolation
 - (D) weathering
51. The distance once around a circle is called its circumference, found by $\pi \times \text{diameter}$. With $\pi \approx 3.14$ and diameter 10 cm, it is about:
- (A) 31.4 cm
 - (B) 10 cm
 - (C) 62.8 cm
 - (D) 78.5 cm
52. The statement 'a number x increased by 7' is written as the expression:
- (A) $x + 7$
 - (B) $7x$
 - (C) $x / 7$
 - (D) $x - 7$
53. A train passes a 1200 m bridge at a steady 20 m/s. The time it takes to cross the length of the bridge is about:
- (A) 24 s
 - (B) 600 s
 - (C) 120 s
 - (D) 60 s
54. Which soil is best suited for growing paddy (rice), a crop that needs water to stand around its roots?
- (A) loose dry soil
 - (B) gravelly soil
 - (C) clayey soil
 - (D) sandy soil
55. A path of uniform width runs inside a square plot. To find the area of just the path, you:
- (A) subtract the inner area from the outer area
 - (B) add the inner area to the outer area
 - (C) divide the outer area by the inner area
 - (D) multiply the two areas together

56. The value of $5y - 4$ when $y = 2$ is:

- (A) 3
- (B) 6
- (C) 14
- (D) 10

57. Two cities are 150 km apart. A car leaves at 9:00 a.m. driving at 50 km/h. It reaches the second city at:

- (A) 10:30 a.m.
- (B) 12:00 noon
- (C) 11:00 a.m.
- (D) 1:00 p.m.

58. A farmer needs 4 kg of topsoil for every 1 square metre of bed. For a bed of area 6 m^2 , the topsoil required is:

- (A) 1.5 kg
- (B) 48 kg
- (C) 10 kg
- (D) 24 kg

59. A rectangular sheet is 8 cm by 6 cm. A square of side 2 cm is cut from one corner. The remaining area is:

- (A) 46 cm^2
- (B) 4 cm^2
- (C) 44 cm^2
- (D) 48 cm^2

60. Which of the following is a constant term - a number with no variable attached?

- (A) $9y$
- (B) $9x$
- (C) $9x^2$
- (D) 9

61. Among these, the SI unit of speed - built from the units of distance and time - is the:

- (A) metre only
- (B) kilometre per hour
- (C) metre per second
- (D) second per metre

62. The amount of water vapour that escapes from soil and plants together returns moisture to the air mainly through:

- (A) weathering of the bedrock
- (B) evaporation and transpiration
- (C) percolation deep underground
- (D) erosion of the topsoil

63. A wire of length 48 cm is bent into a square. The length of each side of the square is:
- (A) 24 cm
 - (B) 48 cm
 - (C) 144 cm
 - (D) 12 cm
64. The product of multiplying $3a$ by $4b$ is:
- (A) $7ab$
 - (B) $12ab$
 - (C) $12a + b$
 - (D) $12a^2b^2$
65. A girl walks once around a rectangular field 60 m long and 40 m wide in 100 s. Her speed (distance = perimeter) is:
- (A) 2.4 m/s
 - (B) 2 m/s
 - (C) 4 m/s
 - (D) 1 m/s
66. The layer of soil lying just below the topsoil, harder and containing less humus, is the:
- (A) subsoil (B-horizon)
 - (B) bedrock (C-horizon)
 - (C) topsoil (A-horizon)
 - (D) humus layer
67. A car wheel of perimeter 2 m makes 100 full turns along a road. The distance the car covers is:
- (A) 2 m
 - (B) 50 m
 - (C) 102 m
 - (D) 200 m
68. A pencil costs $\text{₹}x$ and a pen costs $\text{₹}5$ more than a pencil. The cost of the pen is written as:
- (A) $5 - x$
 - (B) $x - 5$
 - (C) $x + 5$
 - (D) $5x$
69. A motion in which a body retraces its path again and again about a fixed point, like a swing, is called:
- (A) rotational motion
 - (B) uniform motion
 - (C) oscillatory motion
 - (D) rectilinear motion

70. Tiny soil organisms such as earthworms improve the soil chiefly by:

- (A) burrowing to mix and aerate it
- (B) removing all the humus
- (C) turning the soil into solid rock
- (D) packing the soil tightly together

71. Two rectangles have the same area of 24 cm^2 . Rectangle P is 6 cm by 4 cm; rectangle Q is 8 cm by 3 cm. Their perimeters are:

- (A) different (P: 20 cm, Q: 22 cm)
- (B) both impossible to find
- (C) equal, both 24 cm
- (D) equal, both 20 cm

72. An expression with only one term, such as $7xy$, is called a:

- (A) binomial
- (B) trinomial
- (C) monomial
- (D) equation

73. A scooter covers the first 30 km in 1 hour and the next 30 km in 2 hours. Its average speed for the whole trip is:

- (A) 15 km/h
- (B) 30 km/h
- (C) 20 km/h
- (D) 25 km/h

74. Soil holds air in the gaps between its particles. This trapped air is important because it is used by:

- (A) clouds to form rain
- (B) the Sun to make sunlight
- (C) rocks to grow larger
- (D) roots and soil organisms to breathe

75. A floor is 4 m long and 3 m wide. Square tiles of side 1 m are used. The number of tiles needed is:

- (A) 12
- (B) 7
- (C) 14
- (D) 24

76. A pendulum's number of swings is given by $30t$, where t is the time in minutes. In 4 minutes the number of swings is:

- (A) 34
- (B) 30
- (C) 7.5
- (D) 120

77. Before clocks, people used the regularly repeating event of a shadow's movement on a marked plate, called a:

- (A) sand clock
- (B) sundial
- (C) pendulum clock
- (D) water clock

78. A field 50 m long and 30 m wide is to be covered with 2 cm of topsoil. The area to be covered is:

- (A) 80 m²
- (B) 3000 m²
- (C) 160 m²
- (D) 1500 m²

79. The area of a triangle is 24 cm² and its base is 8 cm. Using $\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$, its height is:

- (A) 6 cm
- (B) 12 cm
- (C) 16 cm
- (D) 3 cm

80. Simplifying $8m + 3 - 2m + 5$ by combining like terms gives:

- (A) $6m + 8$
- (B) $14m$
- (C) $6m + 2$
- (D) $10m + 8$

81. An athlete runs 100 m in 10 s, then 100 m more in 20 s. Comparing the two legs, the athlete was faster in the:

- (A) neither - the runner was at rest
- (B) both legs equally
- (C) second 100 m
- (D) first 100 m

82. Over-farming the same land without rest or fresh nutrients gradually makes the soil:

- (A) gain more humus on its own
- (B) lose its fertility
- (C) turn into solid bedrock
- (D) drain water far more slowly

83. Suppose you double every side of a square tile; compared with the first one, the new area becomes:

- (A) four times the original
- (B) eight times the original
- (C) two times the original
- (D) the same as before

84. 'Twice a number n decreased by 3' is correctly written as:

- (A) $3 - 2n$
- (B) $2n + 3$
- (C) $2n - 3$
- (D) $n - 3$

85. A simple pendulum makes 30 complete oscillations in 60 seconds. The time period of the pendulum is:

- (A) 30 s
- (B) 90 s
- (C) 0.5 s
- (D) 2 s

86. The level below the ground at which all the gaps in the soil and rock are completely filled with water is the:

- (A) soil profile
- (B) topsoil
- (C) water table
- (D) humus line

87. A square room has area 49 m^2 . Walking once around its walls (the perimeter), the distance covered is:

- (A) 49 m
- (B) 28 m
- (C) 14 m
- (D) 196 m

88. The total distance covered by walking around a rectangle of length l and breadth b , then evaluated for $l = 10$, $b = 6$, is:

- (A) 32
- (B) 16
- (C) 26
- (D) 60

89. On a distance-time graph, the steeper the sloping line, the:

- (A) longer the time period
- (B) smaller the speed of the body
- (C) body is at rest
- (D) greater the speed of the body

90. Which sequence correctly lists soil particles from the LARGEST to the smallest?

- (A) gravel, sand, silt, clay
- (B) silt, clay, gravel, sand
- (C) clay, silt, sand, gravel
- (D) sand, gravel, clay, silt

91. A field is 100 m long and 60 m wide. Manure is spread at 5 kg per 10 m². The manure needed for the whole field is:

- (A) 320 kg
- (B) 600 kg
- (C) 6000 kg
- (D) 3000 kg

92. The expression $4x^2 + 3x - 7$ is called a trinomial because it has:

- (A) no variables at all
- (B) three unlike terms
- (C) two like terms
- (D) only one term

93. A car shows 5000 km on its odometer at the start of a trip and 5320 km at the end. The distance travelled was:

- (A) 10320 km
- (B) 300 km
- (C) 320 km
- (D) 5320 km

94. A water-holding test shows clayey soil keeps far more water than sandy soil. This is mainly because clay has:

- (A) larger particles with bigger gaps
- (B) smaller particles with tinier gaps
- (C) more gravel mixed into it
- (D) a much warmer temperature

95. The perimeter of an equilateral triangle (all sides equal) with each side 7 cm is:

- (A) 28 cm
- (B) 21 cm
- (C) 49 cm
- (D) 14 cm

96. The value of the expression $3(x + 2)$ when $x = 4$ is:

- (A) 24
- (B) 18
- (C) 9
- (D) 14

97. Walking speeds are most conveniently expressed in km/h, while a snail's creep is better expressed in:

- (A) units of time like hours alone
- (B) smaller units like cm per minute
- (C) units of mass like kilograms
- (D) very large units like km per second

98. Adding manure or compost to a tired field helps the crops mainly because it restores the soil's:

- (A) sand particles
- (B) solid bedrock
- (C) nutrients and humus
- (D) standing water

99. A rectangular plot is 25 m long and 16 m wide. The cost of fencing its boundary at ₹10 per metre is:

- (A) ₹82
- (B) ₹820
- (C) ₹410
- (D) ₹4000

100. If 6 identical bricks have a total mass of 6m kg, then the mass of one brick is:

- (A) m kg
- (B) 6 kg
- (C) 36m kg
- (D) 6m kg